



Research Article

The variation of the East Asian summer monsoon on multi-centennial scale during 8.3–2.9 ka BP[☆]

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ABSTRACT

As a key component of the global climate system, the Asian summer monsoon (ASM) exhibits variability on centennial timescales that remains poorly understood. Based on high-precision U-Th dating and multi-proxy analyses ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$ and Mg/Ca) of stalagmite YK1306 from Yangkou Cave, Chongqing, southwestern China, this study reconstructs a hydroclimate variability for southwestern China during 8.3–2.9 ka BP, and elucidate the spatial and temporal characteristics of centennial-scale climate variability in the East Asian summer monsoon (EASM) region during this period. The multi-proxy records indicated ~500-year cycles of hydroclimatic variability in southwestern China during 8.3–2.9 ka BP, and suggest relatively stable climatic conditions across the EASM region during 6.5–4.8 ka BP. On centennial timescales, hydroclimatic changes in southwestern China are broadly synchronous with those in northern China during the mid-Holocene. Records of Antarctic meltwater, the Intertropical Convergence Zone (ITCZ), and hydroclimatic variations in southwestern China exhibit a common ~500-year periodicity. Model simulations indicate that freshwater input into the Southern Ocean enhances the meridional thermal gradient between the hemispheres, strengthening cross-equatorial flow and driving a northward shift of the ITCZ. Therefore, Antarctic meltwater, by modulating the Atlantic Meridional Overturning Circulation (AMOC) and ITCZ, contributed to increased centennial-scale climate variability during 8.3–6.5 ka BP and 4.8–2.9 ka BP. This study proposes that coupled ocean–atmosphere interactions between the Northern and Southern Hemispheres drive centennial-scale oscillations of the ASM, offering important insights into regional hydro-ecological evolution and societal development at centennial timescales.

1. Introduction

The Asian summer monsoon (ASM) is a vital component of the global monsoon system and regulates regional hydroclimate and sustains the water and food security of billions of people (Mohtadi et al., 2016). It exhibits pronounced regional characteristics and holds significant global

relevance (Wang et al., 2017; Zhang et al., 2022). Instrumental records show that monsoonal precipitation in the East Asian summer monsoon (EASM) region displays pronounced spatial heterogeneity (Ding et al., 2018). On interdecadal timescales, precipitation patterns in the EASM domain often exhibit a dipole pattern (characterized by opposite precipitation anomalies between northern and southern China) or a tripole

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pattern (with opposite precipitation anomalies between the Yangtze River basin and both northern and southern China) (Zhang et al., 2018; Wang et al., 2025a; Xue et al., 2025). However, the brevity of instrumental climate records restricts our capacity to disentangle the relative roles of external forcings and internal variability in shaping long-term hydroclimatic evolution and to investigate the underlying mechanisms operating on multi-centennial timescales. Investigating the variability of the ASM during the Holocene provides critical insights into the mechanisms and trajectories of both past and future climate change. Moreover, the Holocene represents a key period for investigating human–environment interactions and their evolution (Chen et al., 2020).

Centennial-scale climate variability during the Holocene has been widely documented across multiple regions and has been linked to significant societal transformations (Zhang et al., 2008; Jiang et al., 2013; Xu et al., 2014; Kathayat et al., 2017; Kang et al., 2018; Xu et al., 2019a; Chen et al., 2020; Zhang et al., 2021a). The centennial-scale variations in the intensity of the ASM indirectly influenced the succession of ancient dynasties by impacting agricultural yields and social stability (Zhang et al., 2008; Xu et al., 2019a; Chen et al., 2020). Previous studies suggest that El Niño–Southern Oscillation (ENSO) (Zhao et al., 2016; Xu et al., 2019a), volcanic eruptions (Kobashi et al., 2017), North Atlantic subpolar gyre (SPG) (Sicre et al., 2021) influence the changes in the ASM at the centennial scale during the Holocene. North–south shifts of the westerlies may have amplified the impact of North Atlantic ice-rafting events, thereby triggering a weakening of the Atlantic Meridional Overturning Circulation (AMOC) (Ait Brahimi et al., 2019; Herzschuh et al., 2019). In addition, Bakker et al. (2017) proposed that centennial-scale climate variability during the Holocene was regulated by melt-water fluxes from Southern Hemisphere ice sheets. During the transition of the Holocene, a teleconnection between the Southern Ocean and the eastern Pacific was established (Pike et al., 2013). On multi-centennial timescale, increased Antarctic meltwater discharge was associated with La Niña-like conditions (Pike et al., 2013). These fluctuations have been linked to multi-decadal minima in solar activity during the Holocene; however, the mechanisms responsible for amplifying these solar signals remain unclear (Solomina et al., 2015).

In the centennial scale of climate change, the surrounding changes of 400–500 years periodicity have attracted much attention. Proxy evidence from the Gulf of Mexico (planktonic foraminiferal $\delta^{18}\text{O}$) and Jeita Cave in the eastern Mediterranean (stalagmite $\delta^{18}\text{O}$) reveals ~ 500 -year cycles in regional hydroclimate, resembling centennial-scale variability associated with Bond events (Poore et al., 2003; Cheng et al., 2015). Notably, similar ~ 500 -year periodicities have also been observed in solar activity, Greenland surface temperature, and North Atlantic Deep Water (NADW) circulation (Cheng et al., 2015). The origin of the ~ 500 -year cycle in the ASM remains debated. Some studies attribute it to low-latitude ENSO dynamics (Xu et al., 2014; Zhang et al., 2025), while others emphasize the role of AMOC variability (Li et al., 2023) or long-term solar forcing (Park, 2017). In summary, climate oscillations on multi-centennial timescales represent a global signal, and their driving mechanisms are not singular (Askjær et al., 2022; Wang et al., 2025b).

During the transition periods from the early to the middle, and from the middle to the late Holocene, significant meltwater events occurred in Antarctica, but current simulation studies have not taken note of this change (Pike et al., 2013; Ashley et al., 2021). Tan et al. (2020) reported an intensification in the amplitude of ASM variability during the late Holocene. Ice core records from the Tibetan Plateau indicate that freshwater inputs around ~ 8.2 ka BP also coincided with enhanced centennial-scale climate fluctuations (Pang et al., 2024). However, whether the amplification of monsoon variability during the early and late periods of the Holocene was related to Antarctic meltwater remains unknown and warrant further investigation.

In summary, although centennial-scale climate periodicities have been widely documented, the underlying mechanisms driving variability in the ASM system remain inconclusive. Future studies should

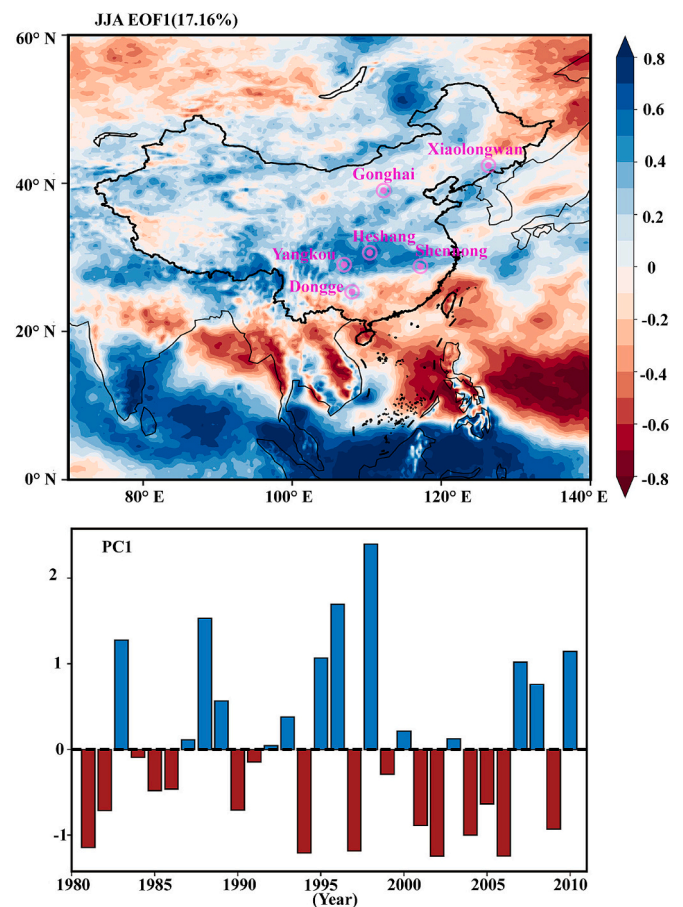


Fig. 1. Locations of the study site. Also shown are the results of Empirical Orthogonal Function (EOF) analysis based on summer precipitation from the ERA5 dataset for the period 1981–2010. The map includes the sites of Dongge Cave (Wang et al., 2005), Heshang Cave (Hu et al., 2008), Shennong Cave (Zhang et al., 2021a), Gonghai Lake (Chen et al., 2015), and Xiaolongwan Lake (Xu et al., 2019a), Yangkou Cave (this study).

emphasize the integration of multiple forcings and feedbacks to better understand the internal dynamics of monsoonal variability. In this study, we employ high-resolution, multi-proxy stalagmite records from Yangkou Cave to reconstruct hydrological changes in southwestern China, providing new insights into the spatial patterns of centennial-scale hydroclimate variability within the EASM domain.

2. Study area and stalagmite sample

The YK1306 stalagmite analyzed in this study was collected from Yangkou Cave, located in Jinfo Mountain, Nanchuan District, Chongqing, southwestern China (Fig. 1). Geologically, Jinfo Mountain lies within the axis of a broad, gentle syncline. The summit is primarily composed of Permian Qixia Formation limestone, which unconformably overlies Silurian sandstones and shales. The middle to lower portions consist mainly of Cambrian dolomite and Ordovician limestone (Zhang et al., 1998). The region experiences a subtropical monsoon climate, predominantly influenced by the Asian monsoon system, with a mean annual temperature of $8.5\text{ }^{\circ}\text{C}$ and an average annual precipitation of approximately 1400 mm (Zhang et al., 1998; Li et al., 2014).

Yangkou Cave ($29^{\circ}02'\text{N}$, $107^{\circ}11'\text{E}$), situated at the summit of Jinfo Mountain at an elevation of 2140 m, extends 2245 m in length (Fig. S1). The cave exhibits a corridor-like platform, with relatively large passage dimensions—typically 15–20 m in width and 8–12 m in height (Li et al., 2014; Chen and Li, 2018). The mean annual temperature within the cave is approximately $7.5\text{ }^{\circ}\text{C}$.

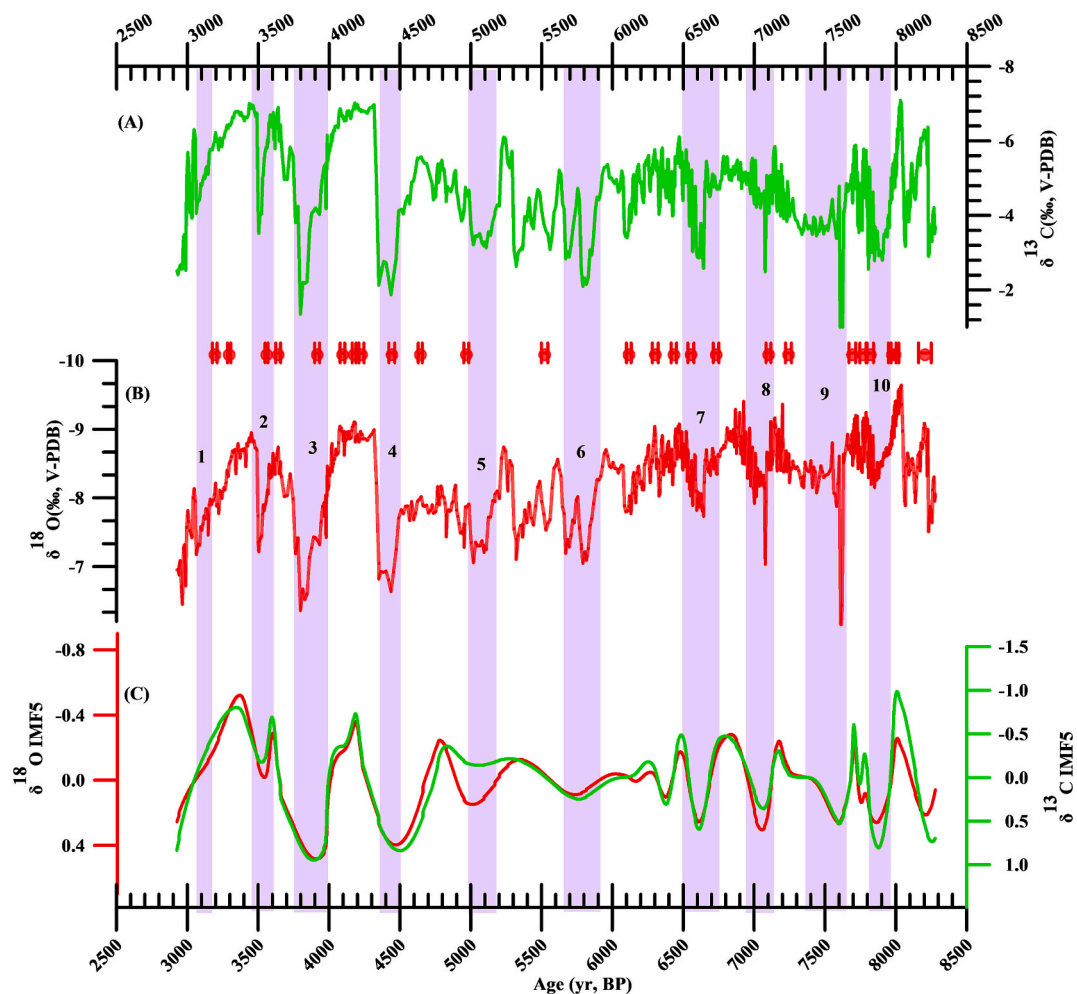


Fig. 2. Variation of $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ in stalagmite YK1306. (A) $\delta^{13}\text{C}$; (B) $\delta^{18}\text{O}$, with red error bars indicating U-Th dating points and associated dating uncertainties; (C) Centennial-scale components (IMF5) of $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ extracted by EEMD analysis. Light purple bands numbered 1 to 10 denote weak ASM events. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The YK1306 stalagmite was collected as a broken specimen during a cave survey. The top portion was missing. The sample was longitudinally sectioned along its growth axis, revealing a brownish profile (Fig. S2). The basal diameter is approximately 80 mm, tapering slightly to 75 mm at the top, with a total length of 210 mm. Although five minor growth axis deviations were observed, the stalagmite is generally cylindrical in shape. Internally, it displays wide, flat laminae (Fig. S2), which facilitated consistent sampling for stable isotopes, trace elements, and U-Th dating along a continuous growth axis.

3. Methods

3.1. X-ray diffraction (XRD) analysis

To assess the mineralogical composition of stalagmite YK1306, seven powder samples were micro-drilled parallel to the growth laminae using a 1 mm-diameter dental drill. Each sample weighed approximately 60 mg and was collected from different depths along the growth axis: 19, 32, 54, 85, 120, 160, and 182 mm from the top of the stalagmite (sampling positions shown in Fig. S2B).

XRD analysis was performed using a TD3500010301 diffractometer at Southwest University. The instrument was operated at 35 kV and 25 mA, with a scanning range of $10\text{--}70^\circ 2\theta$ and an angular resolution of 0.02° . Before analysis, powdered samples were evenly spread on glass slides and then subjected to XRD measurements. The resulting

diffraction patterns were processed and interpreted using MDI Jade 6.5 software.

3.2. U-Th dating

A total of 25 U-Th dating samples were collected along the polished surface of stalagmite YK1306 using a 1 mm-diameter dental drill (Fig. S2B). The analytical procedures followed those described by Cheng et al. (2000). Uranium and thorium were chemically separated following the standard protocols of Edwards et al. (1987). The ^{230}Th dating was performed at the Isotope Laboratory of the Institute for Global Environmental Change, Xi'an Jiaotong University, using a multi-collector inductively coupled plasma mass spectrometer (MC-ICP-MS).

The analytical uncertainties of U and Th isotopic measurements are reported at the 2σ level, and account for corrections related to procedural blanks, background noise, abundance sensitivity, and spike calibration for each isotope. The decay constants used were: $9.1705 \times 10^{-6} \text{ yr}^{-1}$ for ^{230}Th and $2.82206 \times 10^{-6} \text{ yr}^{-1}$ for ^{234}U (Cheng et al., 2013), and $1.55125 \times 10^{-10} \text{ yr}^{-1}$ for ^{238}U (Jaffey et al., 1971).

3.3. Stable isotope and trace element analyses

Stable isotope samples were collected along the central growth axis of stalagmite YK1306 at 0.25 mm intervals using a diameter dental drill (Fig. S2B). These samples were analyzed for stable oxygen and carbon

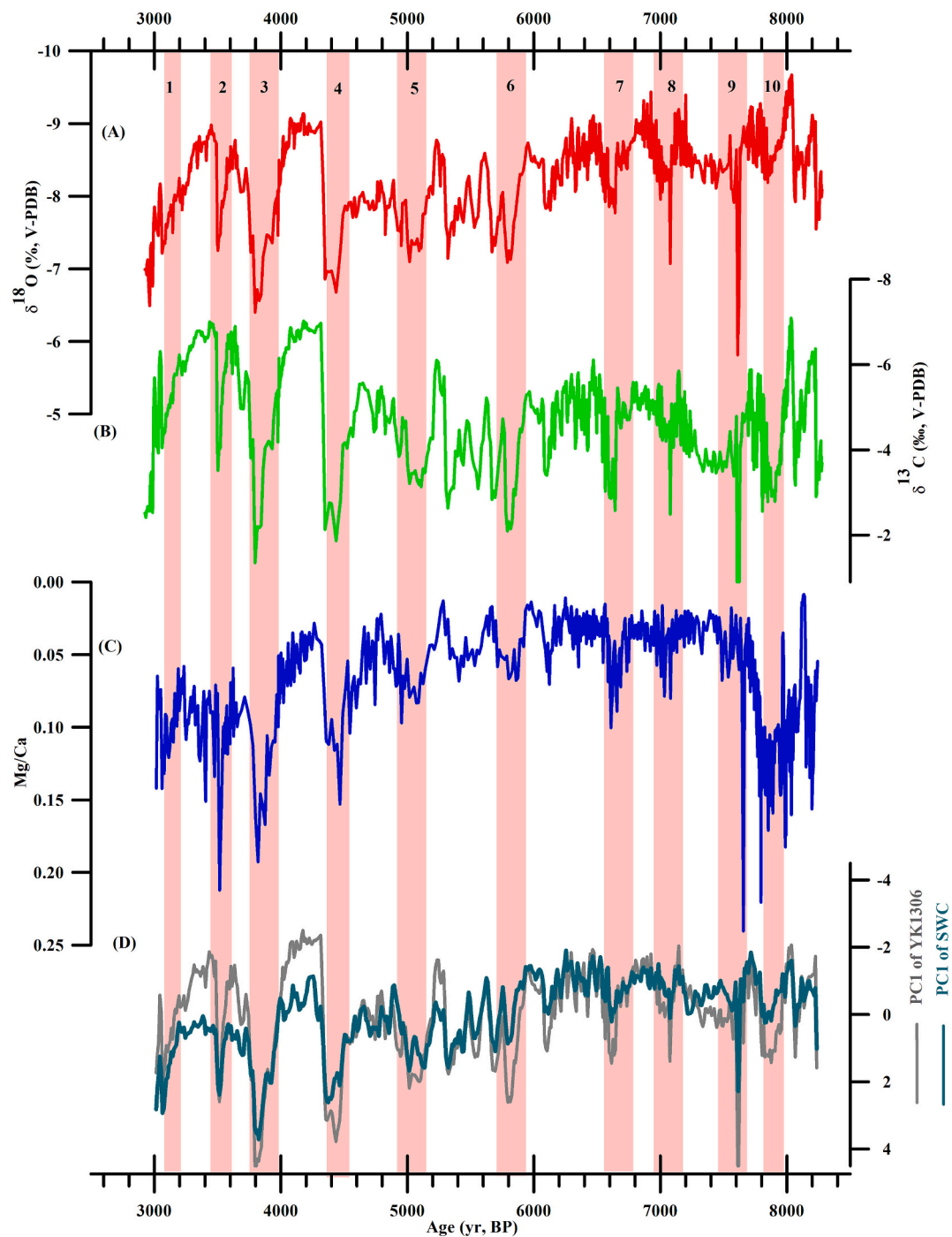


Fig. 3. Multi-proxy comparison of YK1306 stalagmite. (A) $\delta^{18}\text{O}$; (B) $\delta^{13}\text{C}$; (C) Mg/Ca ratio; (D) PC1 of YK1306 denotes the first principal component of the multi-proxy dataset from stalagmite YK1306 (this study); PC1 of SWC represent hydroclimate variability in southwestern China. Pink bands numbered 1 to 10 indicate drought events. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

isotopes ($\delta^{18}\text{O}$ and $\delta^{13}\text{C}$).

Trace element concentrations of YK1306 were determined using a laser-induced breakdown spectrometer (ACCULIBS2500, referred to as LIBS), with measurements taken at 0.3 mm intervals. The analytical procedures for both stable isotopes ($\delta^{18}\text{O}$ and $\delta^{13}\text{C}$) and trace elements (Ca and Mg) followed those described in [Chen et al. \(2025\)](#).

3.4. Statistical methods

Prior to wavelet and cross-spectral analyses, the $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ data from stalagmite YK1306 were linearly interpolated to an evenly spaced

time series at 10-year intervals using OriginPro 2018C. Wavelet transform and cross-wavelet spectrum analyses were then performed using MATLAB R2018b, following the method of [Grinsted et al. \(2004\)](#).

Empirical Mode Decomposition (EMD) is a commonly used technique for analyzing nonlinear and non-stationary time series ([Huang et al., 1998](#)). The Ensemble Empirical Mode Decomposition (EEMD) method, an enhancement of EMD, effectively reduces mode mixing and eliminates transient noise ([Wu and Huang, 2009](#)). EEMD decomposes the original signal into multiple intrinsic mode functions (IMFn, where n denotes the number of modes) and a residual trend component (Res) ([Yang et al., 2017](#)). This approach allows for the identification of

periodicities and amplitudes across different timescales and enables calculation of the relative contribution of each mode to the total variance (Yang et al., 2017). This approach has been widely applied in paleoclimate reconstructions and in studies aimed at separating different signals (Xu et al., 2019a; Tan et al., 2020).

3.5. HadGEM3-GC3.1 model

In order to have a comprehensive understanding the response of the ASM circulation and precipitation change to Antarctic Ice Sheet meltwater, a freshwater hosing experiment conducted by the HadGEM3-GC3.1 model from the Southern Ocean Freshwater Input from Antarctica (SOFIA) Initiative (Swart et al., 2023) was explored in this study. This global environment model consists of coupled atmosphere, ocean, sea ice and land models and has N96 atmosphere resolution (1.875° longitude, 1.25° latitude) and 1° ocean resolution (Kuhlbrodt et al., 2018). In the antwater experiment, a constant and uniform of 0.1 Sv (Sverdrup; $1\text{ Sv} = 1 \times 10^6 \text{ m}^3 \text{ s}^{-1}$) freshwater flux is released around the Antarctic continent within the 100 km against the coast and runs for 100 model years so that the atmosphere and upper ocean reach a semi-equilibrium state, and the climatology of the last 30 years is considered. In addition, a control experiment (piControl) with no additional freshwater anomaly also is used for analysis.

4. Results

4.1. Chronology and mineral of stalagmite YK1306

XRD results indicate consistent peak positions and shapes across all samples (Fig. S3), which were identified as characteristic reflections of calcite. This confirms that the entire stalagmite is composed exclusively of calcite, with no detectable mineralogical variation among the sampled layers.

The U-Th dating results for stalagmite YK1306 are presented in Table S1. The ^{238}U concentrations range from 1502.5 to 25579.1 ppb, with an average of 13104.5 ppb. The mean ^{232}Th concentration is 0.328 ppb. The combination of high ^{238}U and low ^{232}Th concentrations ensures the reliability of precise age determinations for the entire stalagmite. All 25 U-Th ages are consistent with the stratigraphic growth sequence of the stalagmite, showing no evidence of significant depositional hiatuses or age inversions. The average 2σ uncertainty of the U-Th dates is approximately 20 years. Based on these dates, a robust age model was constructed using the ModAge software (Fig. S2A), and was used to establish the temporal framework for the stable isotope and trace element records. The modeling results indicate that YK1306 grew continuously between 8236 yr BP and 2926 yr BP.

4.2. Stable isotopes

The $\delta^{18}\text{O}$ values of stalagmite YK1306 are shown in Fig. 2. The $\delta^{18}\text{O}$ ranges from -9.7‰ to -5.8‰ , with an average value of -8.4‰ . The average temporal resolution of the $\delta^{18}\text{O}$ record is approximately 7 years. Using Ensemble Empirical Mode Decomposition (EEMD), the $\delta^{18}\text{O}$ time series was decomposed into eight Intrinsic Mode Functions (IMFs) and one residual trend component (Res) (Fig. S4 and Fig. 2). The periodicities and variance contributions of each IMF are listed in Table S2.

IMFs 7 and 8 reflect sub-orbital scale variability, accounting for 31 % of the total variance. IMF6 captures millennial-scale variability (13 % contribution), while IMFs 3–5 represent centennial-scale changes (40 % contribution). IMFs 1 and 2 reflect decadal-to-multidecadal variability (16 % contribution). The residual trend shows a gradual increase in $\delta^{18}\text{O}$ values over time, indicating a long-term trend toward heavier isotopic composition.

The $\delta^{13}\text{C}$ values range from -7.1‰ to -0.3‰ , with a mean of -4.9‰ . The $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ values show a statistically significant positive correlation ($R = 0.66$, $n = 830$, $p < 0.01$). On the centennial scale, both

$\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ exhibit coherent variability. During ten identified weak monsoon events, both proxies show simultaneous positive excursions (Fig. 2). Cross-wavelet transform analysis reveals a shared ~ 500 -year periodicity between $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$, with a consistent in-phase relationship (Fig. S5). These results suggest that $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ in YK1306 jointly capture centennial-scale climate variability between 8.2 ka BP and 2.9 ka BP. Based on the centennial-scale oscillations in the $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ records, we identify ten centennial-scale drought events in the period from 8.3 ka BP to 2.9 ka BP (Fig. 2C).

Furthermore, both $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ data indicate a relatively stable phase of the ASM between 6.5 ka BP and 4.8 ka BP. In contrast, the intervals 8.3–6.5 ka BP and 4.8–2.9 ka BP show enhanced centennial-scale variability in ASM intensity (Fig. 2C).

4.3. Trace elements

Stalagmite YK1306 is dense with a smooth and polished surface, which is well-suited for Laser-Induced Breakdown Spectroscopy (LIBS) measurements. To verify the reliability of the results, the sample was analyzed twice using LIBS, and both measurements yielded consistent results with well-matched fluctuation patterns, confirming the reproducibility and robustness of the data (Fig. S6). Due to saturation effects in Sr concentrations within YK1306, the Sr signal exceeded the detection limit of the LIBS system. Therefore, Sr/Ca data are not discussed in this study. The LIBS data used for interpretation are based on the second measurement, which has an average temporal resolution of approximately 8 years. As shown in Fig. 3, during the ten weak monsoon events identified in the $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ records, the Mg/Ca ratio consistently increased, suggesting a coherent geochemical response to changes in hydroclimatic conditions.

5. Discussion

5.1. Multi-proxy comparison of stalagmite YK1306

High-resolution $\delta^{13}\text{C}$, $\delta^{18}\text{O}$, and Mg/Ca records from stalagmite YK1306 exhibit coherent variability at centennial timescales (Fig. 3). According to previous studies, the stalagmite $\delta^{18}\text{O}$ record in ASM region reflects variations in both regional precipitation and atmospheric circulation (Wang et al., 2022; Hu et al., 2025). Both the stalagmite $\delta^{13}\text{C}$ and Mg/Ca records serve as indicators of changes in the regional hydrological environment (Wolf et al., 2023; Chen et al., 2025; Wang et al., 2025b). A more detailed discussion of the interpretation of multi-proxy signals from Yangkou Cave stalagmites is provided by Chen et al. (2025). The co-variability among $\delta^{18}\text{O}$, $\delta^{13}\text{C}$, and Mg/Ca is more indicative of changes in regional hydroclimate conditions.

The $\delta^{18}\text{O}$, $\delta^{13}\text{C}$, and Mg/Ca series were linearly interpolated into evenly spaced 10-year intervals. Correlation analysis shows that all three proxies are significantly correlated with each other (Table S3). Principal component analysis (PCA) can be employed to reduce the dimensionality of multivariate data, thereby identifying common signals across variables while minimizing information loss (Hu et al., 2025). PCA was then applied to these standardized time series, and the first principal component (PC1) explains 61 % of the total variance. PC1 of YK1306 is positively correlated with $\delta^{18}\text{O}$, $\delta^{13}\text{C}$, and Mg/Ca, suggesting that it captures a common hydroclimatic signal. By integrating stalagmite $\delta^{18}\text{O}$ records from southwestern China, Hu et al. (2025) reconstructed a regional precipitation index (PRI). Regional multi-proxy cross-validation may provide a more reliable representation of hydroclimatic variability (Zhao et al., 2021). This study integrates the multi-proxy hydroclimatic signal (PC1) from stalagmite YK1306 with the regional precipitation index (PRI) of southwestern China through principal component analysis (PCA), yielding an PC1 of SWC that explains 57 % of the total variance. Therefore, PC1 of SWC is employed as a representative index of hydroclimate variability in southwestern China (Fig. 3).

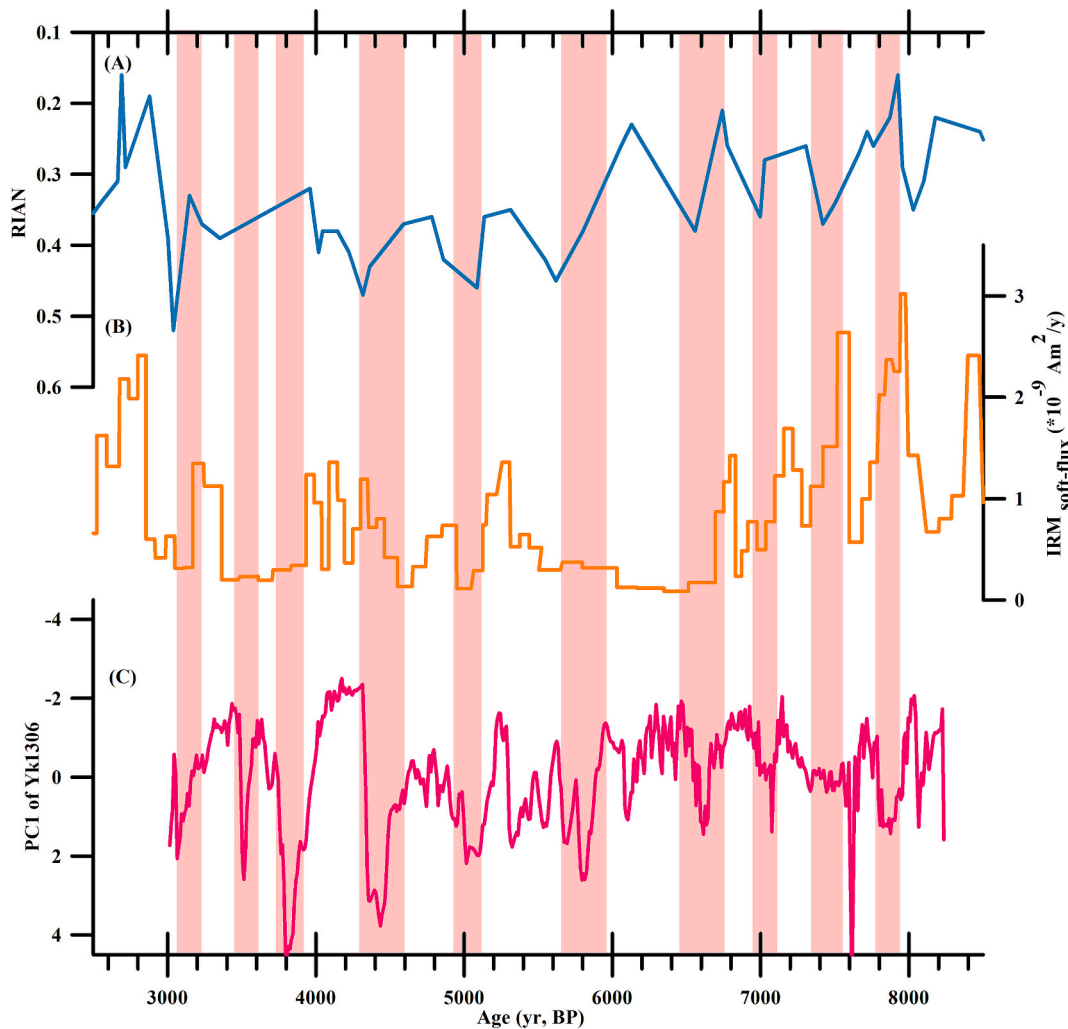


Fig. 4. Comparison of multi-proxy records from adjacent regions. (A) The negative logarithm of Branching Ratio (RIAN) from Heshang Cave stalagmite (Wang et al., 2018); (B) Soft magnetic mineral flux from Heshang Cave stalagmite (Zhu et al., 2017); (C) PC1 of YK1306 (this study). Pink shading indicates drought events. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.2. Centennial-scale climate variability recorded by stalagmites in the adjacent region

The $\delta^{18}\text{O}$ record of stalagmite YK1306 exhibits the overall increasing trend (Fig. S4E), consistent with other stalagmite records from the Asian monsoon region on orbital timescales (Fig. S7), indicating a progressive weakening of the ASM from the Middle to Late Holocene. Compared to other cave $\delta^{18}\text{O}$ records, the YK1306 stalagmite reveals more pronounced centennial-scale weak ASM events. This may be attributed to the high elevation of Yangkou Cave (2140 m), with Jinfo Mountain having a relative height difference of up to 1600 m. As $\delta^{18}\text{O}$ values generally decrease with increasing altitude, their amplitude of variability also tends to be greater (Cai et al., 2012). A similar pattern has been observed in stalagmite records from Tianmen Cave in the southern Tibetan Plateau, where high-altitude settings exhibit larger $\delta^{18}\text{O}$ fluctuations (Cai et al., 2012). A comparison between the $\delta^{18}\text{O}$ records from Yangkou and Tianmen caves reveals significant correlation during overlapping periods ($R = 0.40$, $p < 0.01$) (Fig. S8), underscoring the heightened sensitivity of high-altitude stalagmite records to hydroclimate variability.

Based on stalagmite records from Heshang Cave, Wang et al. (2018) and Zhu et al. (2017) reconstructed regional precipitation variability using two complementary hydroclimatic proxies: the branched index of 3-hydroxy fatty acids (RIAN) and the soft isothermal remanent

magnetization flux ($\text{IRM}_{\text{soft-flux}}$). The RIAN index reflects soil pH influenced by the abundance of Gram-negative bacterial lipids, which shows a negative correlation with precipitation (Wang et al., 2018). $\text{IRM}_{\text{soft-flux}}$ represents the input of soft magnetic minerals, positively correlated with rainfall (Zhu et al., 2017). At the centennial scale, the stalagmite RIAN and $\text{IRM}_{\text{soft-flux}}$ records from nearby regions show a good consistency with the regional hydroclimatic changes reconstructed from multiple proxies of stalagmite YK1306 (Fig. 4).

Furthermore, the $\delta^{18}\text{O}$, $\delta^{13}\text{C}$ and Mg/Ca ratios of stalagmite YK1306 collectively indicate a relatively stable phase of the ASM between 6.5 ka BP and 4.8 ka BP, while enhanced centennial-scale variability occurred during 8.3–6.5 ka BP and 4.8–2.9 ka BP (Fig. 2C). The concordant trend of increased precipitation in the adjacent region during 8.3–6.5 ka BP and 4.8–2.9 ka BP, as recorded by both RIAN and $\text{IRM}_{\text{soft-flux}}$ (Fig. 4), reinforces the hydroclimatic signals derived from the multi-proxy YK1306 stalagmite. The $\delta^{18}\text{O}$ record from Heshang Cave and the $\delta^{13}\text{C}$ record from Luoshui Cave, both located within the Yangtze River Basin, were also subjected to EEMD decomposition to extract the ~500-year periodic components. These components were then compared with those from $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ records of the YK1306 stalagmite (Fig. S9). During the interval from 8.3 ka BP to 2.9 ka BP, the centennial-scale climate variability reflected by the $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ records from Yangkou, Heshang, and Luoshui caves exhibited notable similarities (Fig. S9). Specifically, the climate in southwestern China was relatively stable

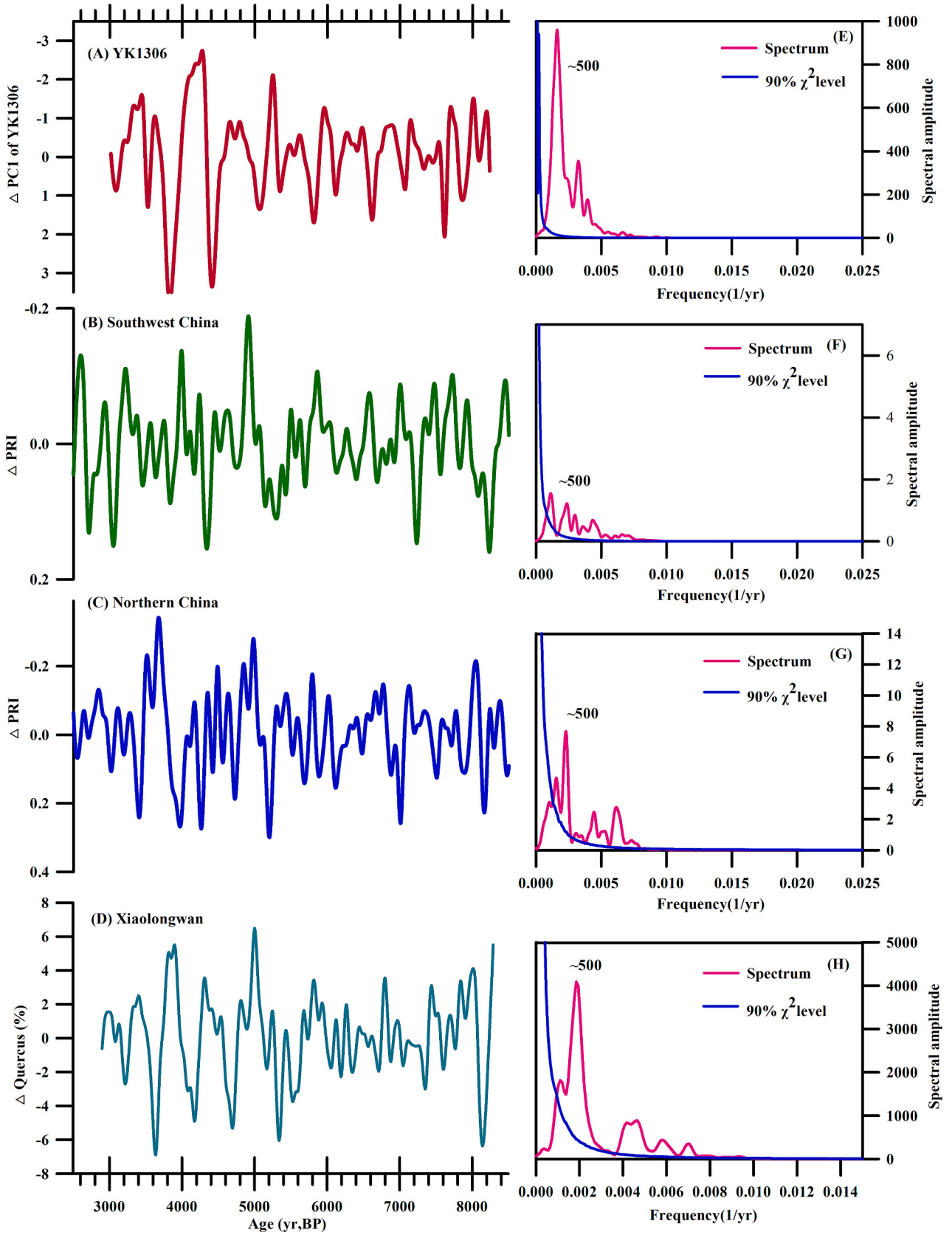


Fig. 5. Comparison of records from the Asian monsoon region on centennial timescales. (A) PC1 of YK1306 (this study); (B) The precipitation index (PRI) of southwestern China (Hu et al., 2025); (C) The precipitation index (PRI) of northern China (Hu et al., 2025); (D) Quercus of Xiaolongwan (Xu et al., 2019a). Panels (A)–(D) show the sum of EEMD components in the 100–1000 year band. Panels (E)–(H) display the corresponding spectral power analyses (Schulz and Mudelsee, 2002).

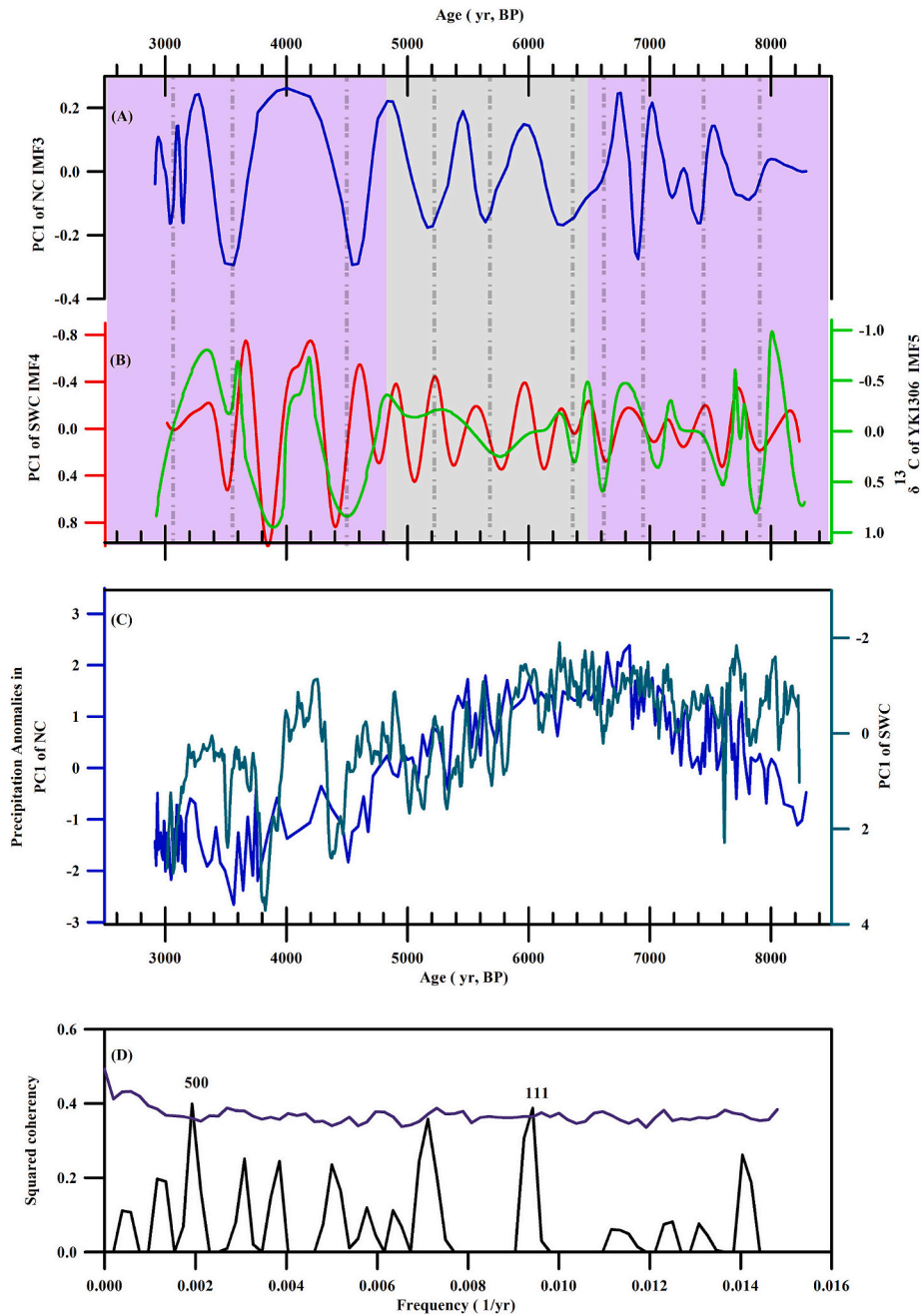


Fig. 6. Comparison of precipitation anomalies between northern China and southwestern China. (A) and (C) Blue lines represent precipitation anomalies in northern China; PC1 of NC (explained variance 76 %) is the first principal component of Gonghai Lake (Chen et al., 2015) and Xiaolongwan Maar Lake records (Xu et al., 2019a), and IMF3 is the centennial-scale component of PC1 of NC; (B) IMF5 is the centennial-scale component of $\delta^{13}\text{C}$ YK1306 stalagmite; IMF4 is the centennial-scale component of PC1 of SWC; (C) PC1 of NC and SWC denote hydroclimatic changes in northern and southwestern China, respectively; (D) Cross-spectral analysis of precipitation anomalies between PC1 of NC (northern China) and PC1 of SWC (southwestern China), the purple lines represent the 90 % confidence level. The methodology follows Björg Ólafsdóttir et al. (2016). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

between 6.5 ka BP and 4.8 ka BP, while increased centennial-scale variability occurred during 8.3–6.5 ka BP and 4.8–2.9 ka BP (Fig. 2 and Fig. S9). A similar pattern is also observed in the $\delta^{18}\text{O}$ records from ice cores on the Tibetan Plateau (Pang et al., 2024). These suggest that the centennial-scale oscillations of the ASM became more pronounced during the early and late stages of the Middle Holocene.

5.3. Spatiotemporal characteristics of centennial-scale hydroclimate variability in the EASM region during the Middle Holocene

Based on reconstructed hydroclimatic indices from different regions, this study conducts a comparative analysis of the spatial distribution characteristics of precipitation variability in the EASM region on centennial timescales. To better highlight the centennial-scale evolution of the ASM, we applied the EEMD method to remove signals at both sub-centennial and supra-millennial timescales (Fig. 5). The filtered records reveal a prominent ~500-year periodicity in centennial-scale monsoon

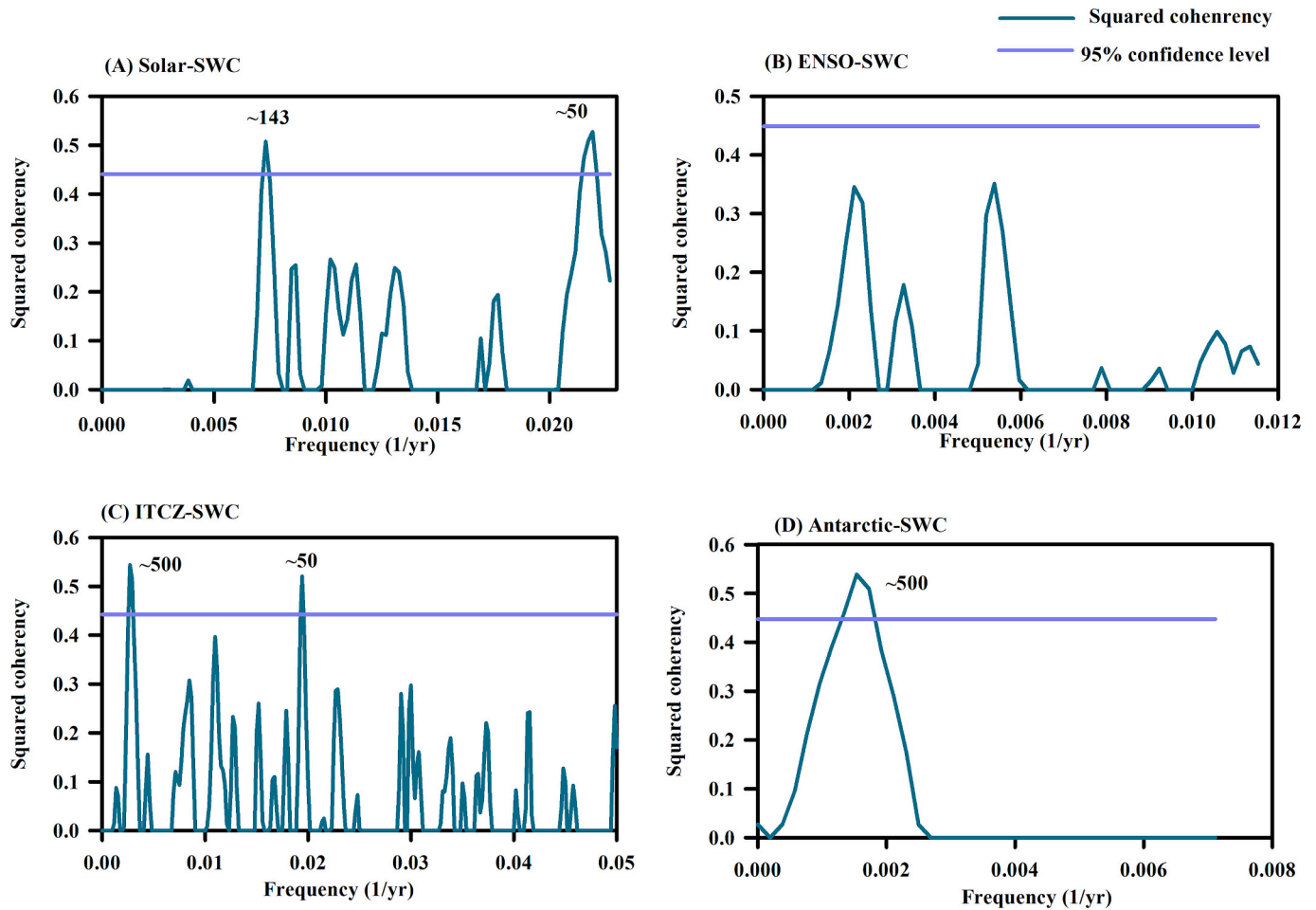


Fig. 7. Cross-spectral analysis. (A) Total solar irradiance (Steinhilber et al., 2012) and PC1 of SWC; (B) ENSO variability (Conroy et al., 2008) and PC1 of SWC; (C) Ti content of the Cariaco Basin (Haug et al., 2001) and PC1 of SWC; (D) $\delta^{18}\text{O}_{\text{diatom}}$ (Pike et al., 2013) and PC1 of SWC. Cross-spectral analysis method follows Björög Ólafsdóttir et al. (2016).

variability during the mid-Holocene across southwest China and northern China (Fig. 5). Similar 400–500-year cycles in monsoon intensity since the last deglaciation have also been identified in stalagmite records from central China and pollen records from northeast China (Xu et al., 2023; Xue et al., 2025).

High-resolution pollen records from the Xiaolongwan Maar Lake in northern China reveal a significant ~500-year periodicity in precipitation (Xu et al., 2019a). Xu et al. (2020) further identified a phase difference in centennial-scale precipitation variability between northern China and the Yangtze River Basin. However, this deduction was based solely on a comparison between two individual sites. To provide a more robust regional assessment, this study integrates precipitation reconstructions from Xiaolongwan and Gonghai Lakes to establish a regional hydroclimate index, better representing centennial-scale precipitation anomalies in northern China (Fig. 6C). Cross-spectral analysis reveals that both southwestern and northern China exhibit a common ~500-year and ~111-year cycle in hydroclimate variability (Fig. 6D). A 100–200-year cycle of climate variability has been well documented in global proxy records and corroborated by modeling studies (Askjær et al., 2022). The precipitation anomalies in the southwestern and northern regions of China tend to show synchronous fluctuations (Fig. 6). Both the integrated analysis of stalagmite records and the TraCE-21 ka simulation results indicate that, between 8 ka BP and 4 ka BP, precipitation variability across the EASM region was largely coherent, except for parts of southeastern China (Hu et al., 2025).

Moreover, instrumental records from 1961 to 2010 also demonstrate a coherent precipitation trend between southwestern and northern

China (Shi et al., 2014). Similarly, the first empirical orthogonal function (EOF1) mode of summer precipitation from 1981 to 2010 reveals a comparable spatial pattern (Fig. 1). The EOF1 analysis of summer precipitation over the past century similarly reveals that southwest and northern China share a coherent pattern of annual precipitation variability (Ding et al., 2018).

On the centennial scale, the pollen record from Xiaolongwan Maar Lake exhibits increased amplitude variability during the periods of 8.3–6.5 ka BP and 4.8–2.9 ka BP (Xu et al., 2019a). Similarly, the integrated precipitation anomaly index for northern China shows comparable oscillatory behavior on the centennial timescale (Fig. 6A). The ~500-year dry density component from Huguangyan Maar Lake further demonstrates that tropical East Asia exhibits comparable oscillatory variability (Liu et al., 2000). This pattern is consistent with the multiproxy reconstruction from the YK1306 stalagmite in southwestern China, further indicating that hydroclimatic variability in southwestern and northern China followed similar trends during the Holocene.

In addition, although both southwestern and northern China exhibited a gradual drying trend around 4.8 ka BP, the drying tendency in southwestern China was relatively weaker (Fig. 6C). Earlier research has likewise shown pronounced regional differences in precipitation between northern and southern China on orbital–equatorial timescales (Zhou et al., 2016; Zhang et al., 2021b; Chen et al., 2024). Nevertheless, while the majority of previous research has concentrated on the Holocene optimum, the spatial variability of precipitation across northern and southern China during the late Holocene at orbital timescales remains insufficiently explored. Hu et al. (2025) also found that, after 4 ka

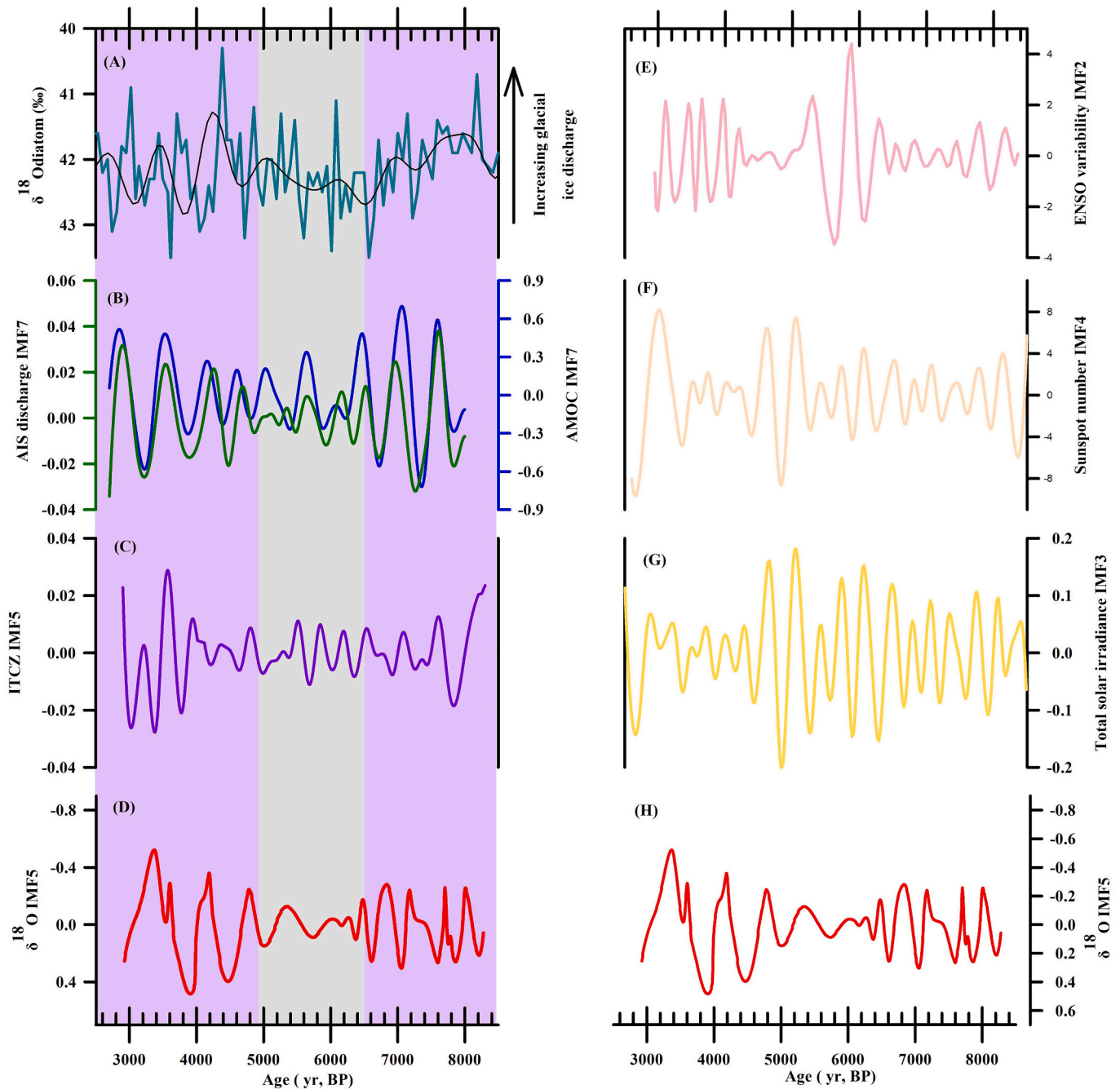


Fig. 8. Comparison of centennial-scale factor components and ASM variability. (A) Glacial discharge along the west Antarctic (Pike et al., 2013); (B) Simulated Antarctic ice-sheet meltwater flux, AMOC strength (Bakker et al., 2017); (C) Ti content of the Cariaco Basin (Haug et al., 2001); (E) ENSO variability (Conroy et al., 2008); (F) Sunspot number (Solanki et al., 2004); (G) Total solar irradiance (Steinhilber et al., 2012); (D) and (H) $\delta^{18}\text{O}$ of YK1306 stalagmite (this study). IMF_s represent EEMD decomposed components (300–500 years). Gray shading indicates more stable climate conditions compared to light purple shading. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

BP, the spatial characteristics of precipitation in the EASM region differed from those of the early to mid-Holocene.

5.4. Centennial-scale influence of Antarctic meltwater and AMOC variability on the ASM region

Previous studies have suggested that solar radiation and ENSO were primary drivers of the ~500-year variability in the EASM (Xu et al., 2014). However, Cross-spectral analysis shows that neither solar activity nor ENSO variability exhibits a significant ~500-year cross-periodicity with the hydroclimatic index from southwestern China (Fig. 7A and

B). Moreover, existing reconstructions indicate that ENSO variability remained relatively stable during the early to middle Holocene and intensified significantly during the late Holocene (Moy et al., 2002; Conroy et al., 2008; Du et al., 2021). Therefore, centennial-scale changes in the ASM are likely modulated by other internal components of the climate system. The latitudinal shifts of the Intertropical Convergence Zone (ITCZ) have been recognized as key factors influencing ASM variability (Wang et al., 2005; Mohtadi et al., 2016; Tan et al., 2019; Xu et al., 2019b). Cross-spectral analysis between the hydroclimatic index from southwestern China and ITCZ migration reveals a common ~500-year periodicity (Fig. 7C), indicating that centennial-scale oscillations of

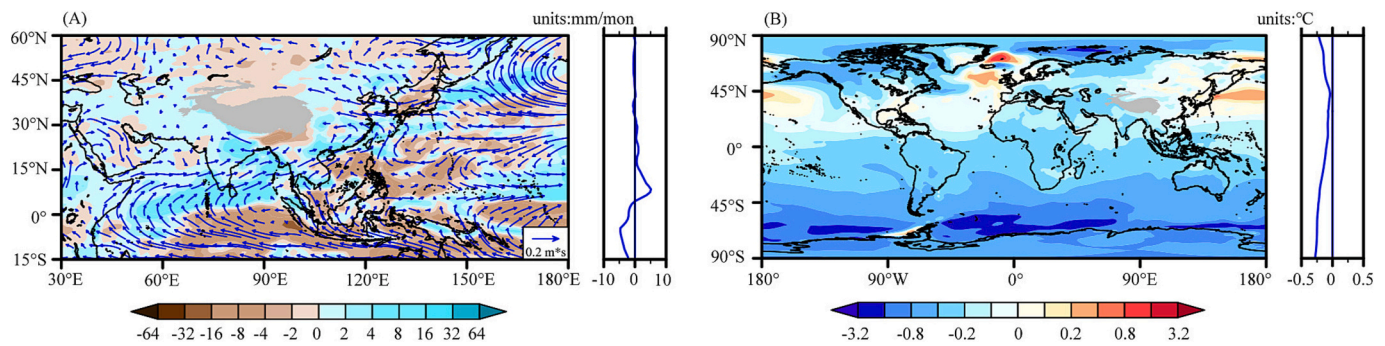


Fig. 9. HadGEM3-GC3.1 model results in response to freshwater injection into the Southern Ocean. (A) Annual precipitation anomalies (shading) and wind field anomalies at 850 hPa (blue arrows), and right panel shows the zonal-mean precipitation anomaly, which defined as the ITCZ position; (B) Global surface temperature anomalies, and right panel shows the zonal-mean surface temperature anomaly. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the ASM during the mid-Holocene were closely linked to shifts in the ITCZ. The meridional shifts of the ITCZ and the subtropical high induced by interhemispheric temperature gradients represent a fundamental mechanism driving precipitation changes in the ASM region (Xu et al., 2019b). In addition, interactions between Antarctic ice sheet meltwater and the AMOC have been shown to amplify internal centennial-scale climate variability during the Holocene (Delworth and Zeng, 2012; Pike et al., 2013; Bakker et al., 2017).

$\delta^{18}\text{O}_{\text{diatom}}$ records from West Antarctic reveal persistent 400–500 years cyclicity in glacial discharge during the Holocene (Pike et al., 2013). Cross-spectral analysis reveals a shared ~ 500 -year periodicity between $\delta^{18}\text{O}_{\text{diatom}}$ and the hydroclimatic index from southwestern China (Fig. 7D). Model simulations indicate that centennial-scale Antarctic meltwater variability corresponds well with ASM changes (Fig. S10) (Bakker et al., 2017). Meltwater-induced changes in the stratification and mixing of Antarctic Bottom Water influence AMOC variability on centennial scales, which in turn affect North Atlantic Sea surface temperature and salinity through teleconnection mechanisms (Bakker et al., 2017; Prange et al., 2023). Salinity anomalies have been identified as a key driver of multi-centennial AMOC variability (Prange et al., 2023). Given that AMOC exhibits intrinsic periodicities in the range of 200–500 years, it is plausible that centennial-scale changes in the ASM are also modulated by Antarctic meltwater dynamics via AMOC-related processes.

Notably, the ASM exhibited a relatively stable phase during 6.5–4.8 ka BP, while centennial-scale variability became more pronounced during 8.3–6.5 ka BP and 4.8–2.9 ka BP (Fig. 8). During 8.3–6.5 ka BP and 4.8–2.9 ka BP, $\delta^{18}\text{O}_{\text{diatom}}$ records from West Antarctica, along with model simulations, indicate enhanced Antarctic meltwater input (Fig. 8A and B). Centennial-scale oscillations of the AMOC and north–south shifts of the ITCZ were also larger than those observed during 6.5–4.8 ka BP (Fig. 8B and C). However, at centennial timescales, variations in ENSO amplitude, solar irradiance, and sunspot numbers do not exhibit similar patterns (Fig. 8E–G). Thus, centennial-scale variability of the AMOC and ITCZ, triggered by Antarctic meltwater flux, likely constitutes an important mechanism driving changes in the ASM.

Model simulations indicate that the injection of Antarctic meltwater leads to a cooling of sea surface temperatures (SSTs) in the Southern Hemisphere and a warming of SSTs in the mid-latitude Northern Hemisphere (Fig. 9B). This enhances the meridional thermal gradient between the hemispheres, intensifies cross-equatorial airflow anomalies (Fig. 9A), and shifts the position of the ITCZ northward (Fig. 9A), ultimately resulting in increased precipitation across most areas of the EASM region (Fig. 9A). The GFDL ESM2M simulations also demonstrate that enhanced Antarctic meltwater input drives a northward shift of the ITCZ, thereby increasing precipitation in the EASM region (Bronselaer et al., 2018). These findings suggest that centennial-scale ASM variability may have been modulated by the coupled Antarctic ice–ocean

system.

6. Conclusions

This study reconstructs a high-resolution hydroclimate variability for southwestern China during 8.3–2.9 ka BP, based on multiple proxies ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$ and Mg/Ca) from the precisely dated YK1306 stalagmite from Yangkou Cave. The agreement among different proxies from geographically proximate sites strongly supports the sensitivity and reliability of the YK1306 record in reflecting regional hydroclimate variability in the adjacent regions. We identify a prominent ~ 500 -year periodicity in the variability of the ASM. Centennial-scale hydroclimatic changes in southwestern China show no significant ~ 500 -year coherence with solar activity or ENSO, yet display a common ~ 500 -year periodicity with Antarctic meltwater flux and ITCZ variability. Principal component analyses of high-resolution pollen records from northern China further reveal that, on centennial timescales, the hydroclimatic variability in southwestern China and northern China exhibits a coherent pattern with a common ~ 500 -year periodicity. The ASM exhibited a relatively stable phase during 6.5–4.8 ka BP. The amplitude of climate variability increased notably during 8.3–6.5 ka BP and 4.8–2.9 ka BP in both regions. Antarctic meltwater may have influenced ASM variability during these intervals by modulating the AMOC and ITCZ, thereby amplifying hydroclimatic fluctuations across the Asian monsoon domain. Model results indicate that following freshwater input into the Southern Ocean, the ITCZ shifts northward and cross-equatorial airflow is enhanced, leading to increased precipitation across most areas of the EASM region.

CRediT authorship contribution statement

Chao-Jun Chen: Writing – original draft, Visualization, Funding acquisition. **Ya-Xi Dong:** Writing – original draft. **Yao Wu:** Visualization, Software, Formal analysis. **Jian Zhang:** Conceptualization. **Hai-Bo Wang:** Investigation. **Yan-Xia Xue:** Conceptualization. **Wen-Jun Sun:** Visualization. **Man-Ping Xie:** Conceptualization. **Hai Cheng:** Methodology, Funding acquisition. **Jun-Yun Li:** Funding acquisition, Formal analysis. **Ming-Qiang Liang:** Methodology. **Ting-Yong Li:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gloplacha.2025.105115>.

Data availability

The origin data are provided in the supplementary material.

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